Uncertainty-Aware Energy Management for Wearable IoT Devices with Conformal Prediction

Dina Hussein*, Chibuike E. Ugwu*, Ganapati Bhat, and Janardhan Rao Doppa
School of Electrical Engineering and Computer Science, Washington State University, Pullman, WA, 99164

2025 62nd ACM/IEEE Design Automation Conference (DAC) | 979-8-3315-0304-8/25/$31.00 ©2025 IEEE | DOI: 10.1109/DAC63849.2025.11132634

***Abstract*—Wearable internet of things (IoT) are transforming diverse healthcare applications including rehabilitation, vital symptom monitoring, and activity recognition. However, the small form-factor of wearable devices constrains the battery capacity and the operating lifetime, thus requiring frequent recharging or battery replacements. Frequent recharging and battery replacement leads to lower quality of service and user satisfaction. Harvesting energy from ambient sources to augment the battery has emerged as an effective solution to improving its operating lifetime. However, ambient energy sources are highly stochastic, making energy management challenging. Prior approaches typically use point predictions for estimating future energy and do not explicitly account for the uncertainty. In strong contrast to prior approaches, this paper presents a conformal prediction-based method for future energy harvest that provides small uncertainty regions with provable coverage guarantees (true output is within the uncertainty region). The uncertainty regions over energy harvest are then leveraged in an energy management algorithm that employs Monte Carlo sampling to evaluate the quality of multiple decisions with varying energy harvests. The decisions are then combined using a lightweight machine learning model to make an energy management decision that is close to the optimal. Experiments on two diverse real-world datasets with about 10 users show that conformal prediction achieves more than 90% coverage with tight prediction intervals; and the energy management algorithm produces decisions that are on average within 2 J of an optimal Oracle, thus showing its effectiveness in improving the quality of service.**

***Index Terms*—Energy Management, Conformal Prediction, IoT Devices**

I. Introduction

Internet of Things (IoT) devices have the potential to enable transformative applications including digital agriculture, health monitoring, wide area monitoring, and smart homes [1–6]. Energy harvesting and management are promising technologies to improve the self-sustainability of IoT devices [7–9]. However, ambient energy sources are highly stochastic with wide variations across locations, time, seasons, and user behavior [7]. Similarly, energy demand from the applications can vary across time due to sensing requirements, user activities, and environmental conditions. For instance, sensors need higher sampling rates and power when a user is performing exercise when compared to sleeping. Accounting for these variations at runtime is crucial to ensure sustainability of the device.

Achieving self-sustainability in IoT devices consists of two key steps: 1) future energy availability prediction and 2) management of the device energy. Future energy availability predictions are important to understand the variability of energy resource in the system. Accurate predictions ensure that the system is able to maintain sufficient energy reserves and does not go into low power mode. The energy management (EM) algorithms must consider the uncertainty in predicted future energy dynamics to make optimal decisions to achieve self-sustainability and maximize application quality [10, 11]. To this end, this paper develops novel approaches for energy predictions with *correct-by-design uncertainty bounds* and *uncertainty-aware* EM decisions.

Energy harvest (EH) predictor must satisfy two main conditions: 1) it must provide high prediction accuracy with respect to ground-truth; and 2) it must produce uncertainty bounds for each prediction with guaranteed high coverage (true value lies within the interval) and small prediction intervals (tight uncertainty bounds). To satisfy these conditions by design, we propose a multi-target conformal prediction (MTCP) approach to obtain small uncertainty regions with coverage guarantees. The key idea behind MTCP is to extend the idea of single target conformal prediction [12] to multiple target variables (one for each future EH interval) and construct a joint uncertainty region. MTCP is a simple and low-overhead approach that allows us to easily incorporate domain constraints (e.g., energy is not negative). By considering multiple target variables, the MTCP approach is able to provide high-quality prediction intervals that adapt to temporal variations in EH data.

Prediction of future EH is not sufficient for sustainable operation of IoT devices. This is because the predictions and uncertainty bounds must be used within the EM algorithm to obtain energy allocation or budget for each decision interval. The energy allocation must account for variations in ambient energy and device energy requirements. To this end, we formulate a constrained optimization problem to obtain energy allocation bounds that satisfy application requirements, maximize application quality, and maintain energy sustainability while accounting for uncertainty in EH.

The EM algorithm employs Monte Carlo sampling under overhead constraints to evaluate multiple EM decisions as per varying EH predictions and associated uncertainty. The sampling is done such that it covers the entire uncertainty region while not exceeding the overhead budget for EM. The EM decisions must be combined to obtain a single energy budget that is as close to an optimal decision made by an Oracle that has access to ground-truth EH values. To this end, we use a lightweight machine learning (ML) model that takes candidate decisions from the Monto Carlo sampling and provides a decision that aims to mimic the Oracle for each interval. In summary, the proposed approach first predicts uncertainty bounds for EH using MTCP and then performs Monte Carlo sampling using EH uncertainty bounds to obtain EM decisions.

We validate the proposed approach using EH and activity data from two diverse real-world datasets [13, 14]. Activity information from the two datasets is combined with solar irradiation data for outdoor and indoor conditions to obtain ground truth EH data. The data are then used to validate the proposed MTCP for EH uncertainty quantification and the uncertainty-aware EM approach. The evaluation shows that the proposed approach produces decisions that are on average within 2 J of an optimal Oracle. End-to-end experiments on a prototype IoT device show that the approach consumes at most 88 mJ energy in the worst-case scenario, with lower consumption under typical conditions while improving the performance by at least 25% compared to baselines.

In summary, this paper makes the following contributions:

- A novel multi-target conformal prediction (MTCP) algorithm that provides guaranteed coverage and tight uncertainty bounds

*Dina Hussein and Chibuike E. Ugwu contributed equally to this work.

for future EH over multiple time horizons. To the best of our knowledge, this is the first approach that has applied MTCP in the context of EH prediction.

- Novel EM algorithm that leverages Monte Carlo sampling to account for uncertainty in the EH to make decisions that are on average within 2 J of an optimal Oracle. The Monte Carlo sampling allows us to smartly trade-off the overhead of EM while accounting for uncertainty in the EH.
- Comprehensive evaluations using two real-world activity datasets with indoor and outdoor EH values. The evaluations show that the proposed approach achieves better performance that at least 25% higher than baseline.

II. Related Work

Accurate prediction of future EH is vital for optimal energy allocation. Previous studies have applied analytical and ML methods to address this task [9, 15–18], primarily focusing on generating single-point predictions of future EH. While these approaches provide estimates, they lack uncertainty quantification, which is essential for reliable EM decisions in wearable IoT systems.

Recent research has explored ensemble methods to construct reliable prediction intervals by combining the mean (μ) and variance (σ) estimators to form prediction intervals [19, 20]. However, these intervals are often overly conservative and frequently fail to guarantee the necessary coverage which is critical for real-world EM applications. Thus, there remains a need for methods that provide tight uncertainty regions while providing coverage guarantees.

We propose to utilize conformal prediction (CP) to overcome the limitations of prior EH prediction approaches, particularly for uncertainty quantification [21]. CP is a distribution-free uncertainty quantification framework for constructing prediction intervals with guaranteed finite-sample coverage for any given predictive model [12, 22–31]. CP constructs prediction intervals by defining a non-conformity score between predictions and ground truth, computing scores on calibration data, and determining a quantile threshold based on a user-defined error rate (say 5%). Motivated by the need for uncertainty regions in EH predictions, this work explores the under-studied application of CP for multi-target regression.

EM for wearable IoT devices is a well-studied problem [7, 8, 32–37]. Prior approaches for EM include linear programming, model predictive control, and reinforcement learning. These approaches aim to perform EM such that the device performance is maximized while ensuring long-term operation. However, prior approaches fail to consider the uncertainty in future EH when making decisions. Consequently, they are unable to react to sudden changes in EH behavior (e.g., cloudy days followed by sunny days). This leads to sub-optimal performance for the device. The proposed work aims to precisely overcome the limitations of prior approaches by reasoning about uncertainty in multiple future intervals, thus making it more robust when compared to baseline approaches.

III. Background and Problem Setup

We consider a wearable IoT system that consists of multiple sensors and harvesters, as shown in Figure 1. We assume that the device integrates a rechargeable battery. We also assume that wearable devices in this work integrate EH from body motion and ambient light to augment the battery.

The goal of EH is to direct energy to the battery for future usage thereby minimizing the frequency of recharging operations. We can potentially eliminate recharging needs if the energy consumed in a given horizon, such as a day, is equal to the harvested energy.

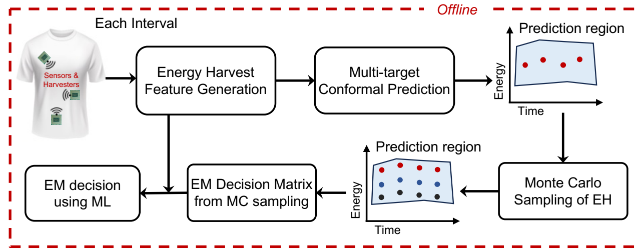

Fig. 1: Overview of our uncertainty-aware energy management approach.

This will lead to energy-neutral operation (ENO) and significantly reduce recharging needs. However, achieving ENO is challenging in practice due to the following reasons.

Uncertainty in EH: Ambient sources of energy are highly stochastic in nature due to variations in lighting conditions, user activity levels, and location. We must quantify the uncertainty in the EH to avoid over- or under-allocation of energy.

Uncertainty-Aware EM: EM algorithms must carefully determine the energy budget in each time horizon to balance two objectives: 1) the maximization of application accuracy and quality of service, and 2) minimizing energy consumption and maintaining sufficient battery levels at all time horizons to successfully maintain ENO. Achieving the two objectives is challenging because the EM algorithm must reason about uncertainty in future EH so that battery energy reserves are effectively managed and not exhausted.

The proposed approaches in this paper precisely aim to handle the above challenges by formulating and solving the following problems for uncertainty quantification and EM.

A. Uncertainty Quantification for EH

We consider that the EM decisions are made with a fixed time horizon of one day with each day divided into T equal length intervals [38]. Our goal is to design a function f that takes a set of input features X and provides *multiple* EH predictions for intervals t to $t+H$, where H is the number of future time intervals. The function f must also provide accurate uncertainty bounds that meet user-specified coverage guarantees at runtime.

B. Energy Management Policy

The goal of the EM policy π is to take EH predictions from f as input and provide an energy budget for interval t that maximizes application quality and energy-sustainability. The policy π must make decisions that also account for the uncertainty in the EH by intelligently sampling the uncertainty. To this end, our goal is to design a policy π that intelligently samples the uncertainty regions in EH while maintaining ENO and maximizing application quality.

IV. Uncertainty Quantification for Energy Harvest Prediction

CP [12, 22–25] is a robust framework that quantifies uncertainty and provides rigorous coverage guarantees for any predictive black-box model. Given a predictive model f, n calibration examples with p features and d targets $\{X_i, Y_i\}_{i=1}^{n}$: $X_i \in \mathbb{R}^p$ and $Y_i \in \mathbb{R}^d$ (which were not used during training), a user-specified error rate α (say 5%), a scoring function $S(X_i, Y_i)$ that measures the discrepancy between the predicted value $f(X_i)$ and the ground truth Y_i (e.g., absolute residual), and a new test input X_{n+1}, instead of a conventional single point prediction, CP generates a prediction set $\mathcal{R}(X_{n+1}) \subseteq \mathbb{R}^d$ that confidently ensures that the true output Y_{n+1} falls within it with a probability of at least $1-\alpha$ on average.

CP can be used to construct prediction sets for classification tasks or prediction intervals for regression tasks. In this work, we focus on the regression setting with multiple target variables, as our goal is to quantify the prediction uncertainty for EH across a prediction horizon $H > 1$ time periods. For a univariate target ($Y_i \in \mathbb{R}^H$) with $H = 1$, the uncertainty region $\mathcal{R}(X) \subseteq \mathbb{R}$ is represented as an interval defined by its lower and upper bounds. However, for multi-dimensional targets ($H > 1$), the uncertainty region $\mathcal{R}(X) \subseteq \mathbb{R}^H$ becomes an H-dimensional region, capturing the joint uncertainty.

A. Multi-Target Conformal Prediction (MTCP)

We propose a lightweight MTCP approach designed to construct uncertainty regions for EH with guaranteed theoretical and empirical coverage with low overhead suited for resource-constrained devices.

Given a user-specified error rate, α, and a prediction horizon H, MTCP employs Conformalized Quantile Regression (CQR) [12] to construct base prediction intervals for each target horizon $h \in \{1, \ldots, H\}$. The adjusted error rate for each target is defined as $\beta = \alpha/H$. The prediction interval for each target is constructed as:

$$\mathcal{R}_h(X) = [\hat{q}^h_{\beta/2}(X), \hat{q}^h_{1-\beta/2}(X)],$$

where $\hat{q}^h_{\beta/2}(X)$ and $\hat{q}^h_{1-\beta/2}(X)$ represent the lower and upper quantile estimates for the h-th target variable, respectively.

To reduce the uncertainty regions, we incorporate the known non-negativity constraint of EH (energy is not negative) into MTCP. Specifically, the prediction interval for each target variable is adjusted to fall in the positive region:

$$\mathcal{R}^{+}_h(X) = \left[\max\{0, \hat{q}^h_{\beta/2}(X)\}, \hat{q}^h_{1-\beta/2}(X)\right],$$

This constraint reduces the size of the uncertainty regions without compromising the coverage guarantees. The final constrained H-dimensional uncertainty region becomes:

$$\mathcal{R}^{+}(X) = \bigcap_{h=1}^{H} \mathcal{R}^{+}_h(X)$$

For a new test input X_{n+1}, the coverage inequality in Equation 1 is guaranteed to hold marginally:

$$\mathbb{P}[Y_{n+1} \in \mathcal{R}^{+}(X_{n+1})] \geq 1 - \beta H \tag{1}$$

where $\mathcal{R}^{+}(X) \subseteq \mathbb{R}^H$ and $\beta H = \alpha$. By constructing uncertainty regions in this manner, the MTCP approach ensures that the true EH values lie within the predicted uncertainty region with a high probability of at least $(1-\alpha)$. This method provides tighter uncertainty regions and remains computationally lightweight, making it well-suited real-time EM.

V. Uncertainty-Aware Energy Management

A. Problem Formulation

We employ a common formulation used to achieve ENO in energy-harvesting devices, first introduced by Kansal et al. [7]. The optimization problem can be formulated as follows:

$$\max. \quad Q(\mathbf{E_A}) = \sum_{t=0}^{T-1} \beta^t \ln\left(\frac{E^t_A}{M_E}\right) \quad \text{s. t.} \tag{2}$$

$$E^{t+1}_B = E^t_B + \eta\mathcal{E}^t_H - E^t_A - E_o, \; 0 \leq t \leq T-1 \tag{3}$$

$$E^{t+1}_B \geq E_{min} \quad 0 \leq t \leq T-1 \tag{4}$$

$$E^T_B \geq E_{target} \tag{5}$$

Objective: The key objective is to maximize the quality of service Q represented by the sum of logarithms of the energy allocation (budget) E^t_A in each interval t. Intuitively, higher energy allocations lead to higher quality of service since devices can have higher sampling rates and run more complex algorithms. The objective is scaled by a constant M_E to ensure that the objective Q is positive only when the device is able to perform a minimum level of sampling and processing for the application.

Constraints: The problem has three main constraints. First, the constraint in Equation 3 specifies that the battery level at the beginning of an interval $t+1$ is governed by the battery in the previous interval t, stochastic EH $\mathcal{E}^t_H$ with efficiency η, energy allocation E^t_A, and overhead of EM decision-making E_o. The second constraint ensures that battery levels in the device are always greater than a minimum value of E_{min}. The E_{min} reserve ensures additional energy for the device when there are mispredictions in EH or incorrect EM decisions. Finally, the constraint in Equation 5 specifies that the battery at the end of the horizon, i.e. E^{t+1}_B is greater than a target value of E_{target}. Consequently, if we ensure that the battery energy at the beginning and end of a horizon is greater than E_{target}, the device will achieve ENO.

Solution Approach: The EM problem is a convex optimization problem with a concave objective and linear constraints [39]. The optimization problem can be solved using convex optimization solvers using interior-point methods [39] or with iterative solvers. Iterative solvers are attractive since their execution can be fine-tuned to balance accuracy and overhead. Inspired by this, we propose to utilize the iterative gradient projection (IGP) algorithm proposed in [40] to solve the optimization problem.

Even with an iterative algorithm, it is challenging to solve the EM problem optimally at runtime due to the unavailability of future EH values. At the same time, we can obtain the upper bound on the quality of EM decisions by solving the problem *offline* with actual EH values. Indeed, we utilize the optimal solutions with actual EH values as the **Oracle** policy to compare the effectiveness of the proposed and baseline EM policies.

B. Monte Carlo Sampling of EH Uncertainty Regions

The uncertainty in EH values must be handled carefully to make decisions that maximize the QoS while maintaining ENO. Algorithm 1 shows the proposed approach to handle the uncertainty in future EH. At a given interval t, the algorithm takes EH predictions and uncertainty regions from MTCP, overhead budget for EM, and current battery status as inputs. The uncertainty bounds provide a region of possible values for EH in the next H intervals. Intuitively, the region is a H dimensional hyperrectangle as explained in the previous section. We must sample this multi-dimensional space to evaluate various EM decision choices. To this end, the primary loop in the algorithm randomly samples an H-dimensional vector from the EH uncertainty region. It is critical to sample the prediction region uniformly to ensure that the EM decisions are not skewed by low or high EH predictions. To this end, we utilize Monte Carlo sampling to obtain random EH prediction samples while maintaining a wide coverage across the uncertainty regions.

The H-dimensional vector is then used as the EH prediction for intervals t to $t+H$ before utilizing IGP to obtain the EM decisions for intervals t to $T-1$. That is, we solve the EM problem from interval t until the end of the current horizon. The decisions are stored in a matrix for further analysis before making the final EM decision. The process is repeated until the energy budget for making EM decisions is exhausted. At the end of iterations, the matrix with

Algorithm 1: Monte Carlo Sampling and EM Decisions

Input: MTCP EH uncertainty region for intervals t to $t+H$, Overhead budget E_o, Features X and Battery E_B^t
$M_{E_A} \leftarrow$ Initialize decision matrix
while $E_o \geq 0$ **do**
  $[\hat{E}_H^t, \ldots, \hat{E}_H^{t+H}] \leftarrow$ Randomly sample EH
  $[\tilde{E}_A^t, \ldots, \tilde{E}_T^{T-1}] \leftarrow$ Solve EM problem with EH samples
  $M_{E_A} \leftarrow [M_{E_A}; [\tilde{E}_A^t, \ldots, \tilde{E}_T^{T-1}]]$
  $E_o \leftarrow E_o - E_{IGP}$
end
Use M_{E_A} and X as input for ML model to obtain EM decision
return EM decision

decisions from each EH sampled vector is provided as output of the algorithm for final decision-making.

C. Lightweight ML Model for Oracle Tracking

The wearable IoT device must make a single decision for the energy budget based on the decisions provided by Monte Carlo sampling. A straightforward approach is to randomly choose a decision among the decisions provided by the sampling. However, the random decision does not consider the optimality of the decisions and closeness to the optimal Oracle. To this end, we propose to leverage the trajectory of decisions provided by the Monte Carlo sampling and MTCP features to train an ML model that aims to obtain decisions that are close to the Oracle. Key components and insights behind the ML model are described in the following.

Input Features: The trajectory evaluations performed by the Monte Carlo sampling include potential decisions for intervals t to $T-1$. Each row of the evaluations considers different levels of uncertainty and we can use the average of these decisions as a feature for the ML model. In particular, the average of decisions allows the feature space to be of the same size when the variable number of samples are obtained in the previous section. The features used for predicting EH values additionally provide insight into past EH values, therefore they can aid in making the final EM decision. Consequently, the feature set for EM decision-making consists of the EH prediction features X and $\hat{E}_A^{t+1} - \hat{E}_A^{t+H}$, where $\hat{E}_A^{t+1}$ represents the average of potential EM decisions.

Training Target: Our goal is to obtain decisions that are close to the Oracle without full system knowledge of the future. Therefore, we use the Oracle EM decision as the training target for the ML model. Specifically, we obtain the Oracle solution offline and use it as the target for the ML model.

Model Structure: Any supervised learning algorithm can be utilized to obtain the EM decisions. In this work, we employ fully connected neural networks due to their simplicity and strong performance in diverse scenarios. Additional details on the neural network structure and its configuration are provided in Section VI-A5.

VI. Experimental Results

A. Experimental Setup

1) Wearable IoT Device Model: Our system design focuses on a wearable IoT device that is equipped with five energy harvesters and multiple sensors. It includes piezoelectric sensors strategically placed at four locations on the knees and elbows to efficiently harvest motion energy. Additionally, the device integrates an SP3-37 [41] flexible photovoltaic (PV) cell for light and solar EH.

2) Datasets: The approach aims to deliver uncertainty-aware predictions of energy harvest in both indoor and outdoor environments. Therefore, we leverage activity data from two publicly available datasets: 1) ARAS [13] and 2) Mannheim [14]. The ARAS dataset includes activity traces collected over 30 days from four users residing in two houses, while the Mannheim dataset provides data from six users over a two-week period.

The ARAS and Mannheim datasets provide user activity data spanning one month and two weeks, respectively. While useful, these durations are insufficient for our goal of evaluating accuracy across different seasons and times of the year. To address this limitation, we extend both datasets to cover a six-year period (2015–2020). This is achieved by shuffling and augmenting the original activity data, allowing us to simulate long-term usage patterns and conduct comprehensive evaluations. The activity data are combined with typical EH values from motion and ambient light irradiance to obtain actual EH values in each interval [9, 42].

3) Data Splitting: The six-year energy dataset is divided into three subsets: 1) training, 2) validation, and 3) testing. The data from 2015 is allocated for training, 2016 is reserved for validation, and remaining data are used for testing.

4) Proposed CP Parameters: The parameters for MTCP consist of the base CQR prediction model architecture, prediction horizon H, and error rate $\alpha = 0.1$ corresponding to a 0.9 expected marginal coverage level. The model architecture is three layers and 64 neurons per layer network. Given prediction horizon $H \in \{2, 3, 4, 5\}$, for each target $h \in \{1, ..., H\}$, the error rate is set to $\beta = 0.1/H$.

5) Proposed EM Parameters: The proposed EM framework parameters consist of the decision intervals, battery constraints, and ML model structure. The EM horizon spans 24 hours, divided into 24 decision intervals throughout the day. The battery energy target is set to 100 J, with a minimum energy threshold of 10 J. The framework incorporates a discount factor γ of 0.99. The utility ME is defined as eight. The ML model used in the EM framework consists of four fully connected layers with 32 neurons per layer.

6) Evaluation Metrics: The MTCP framework is evaluated for the coverage and size of the uncertainty region. Next, we evaluate the quality of the individual EM decisions by comparing it to the optimal Oracle that uses actual EH values. We use the mean absolute error (MAE) compared to the Oracle to evaluate the proposed EM approach and baseline algorithms. Finally, the cumulative QoS Q over the horizon is used to evaluate application performance.

B. Baseline Methods for Comparison

1) Baseline Point Prediction Method: The most common approach to perform EM is to use point predictions of EH to make decisions using iterative algorithms. While any point prediction approach is applicable, we utilize the average of last three days of actual EH values as the point prediction in this approach due to its simplicity and ease of computation.

2) Baseline uncertainty region with Ensemble Method: The ensemble method uses a collection of random forests to first obtain EH predictions for each interval. The individual predictions are then combined to form prediction intervals by using the mean (μ) and variance (σ) of predictions. Specifically, we can obtain the bounds as: $(\mu \pm \sigma)$. The final uncertainty region for all horizons is obtained as the H-dimensional intersection of all prediction intervals.

3) Random Choice Energy Decision (MTCP-Random): One of the straightforward methods is to choose the final EM decision randomly from the range of decisions provided by the Monte Carlo sampling. To this end, the MTCP-Random serves as a baseline

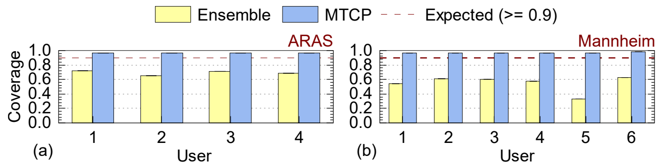

Fig. 2: Coverage comparison between Ensemble and MTCP prediction uncertainty regions for ARAS and Mannheim datasets across all users in Horizon 4. Similar trends are observed across all other prediction horizons.

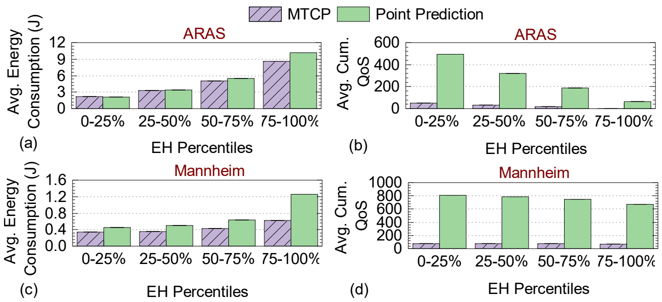

Fig. 3: Comparison of average energy consumption and absolute of average cumulative utility of MTCP, and point prediction with user one in ARAS and Mannheim datasets. Lower cumulative values lead to better quality since the logarithms are typically negative.

by randomly choosing an energy allocation in the range provided by the sampling. We use this baseline to highlight the benefits of incorporating ML-based decision-making in the proposed method.

C. Evaluation of the Proposed MTCP Approach

A critical aspect of predicting future EH is the ability to manage the prediction error effectively and to quantify the uncertainty associated with predictions. Achieving a coverage level that meets or exceeds the user-specified expectation is essential for reliable and accurate future energy allocation. To this end, this section evaluates the coverage with different prediction horizons H. Figure 2 shows the coverage for the proposed MTCP and ensemble methods. Figure 2 shows that the ensemble method fails to achieve the desired coverage level of 0.9, while the proposed MTCP approach exceeds the desired coverage for all users in both datasets, thus showing the effectiveness of our approach.

D. Evaluation of the Proposed EM Approach

1) Energy Management with Point Predictions: We compare the performance of the proposed approach with point prediction. Figure 3 compares the average energy allocation and the absolute value of the average cumulative utility for both approaches using data from user one in the ARAS and Mannheim datasets. Each bar represents a range of EH, allowing for an analysis of trends across low to high-energy availability scenarios. The results highlight key energy efficiency and QoS differences between the proposed approach and the point prediction approach. Specifically, the proposed approach demonstrates improved quality consistency across percentiles compared to point prediction. Specifically, the point prediction approach has worst utility even when consuming more energy due to its inability to account for uncertainty, leading to overallocation and battery depletion. Overall, these results demonstrate the importance of considering uncertainty EM.

2) Validation with Different Prediction Horizons: Next, we evaluate the proposed approach with different prediction horizons given as input to the CP algorithm. Specifically, we vary the prediction horizon H from two to five and evaluate the EM performance.

TABLE I: Mean of MAE for MTCP, MTCP-Random, Ensemble-ML, Ensemble-Random, and Point Prediction methods with different prediction horizons across years for user 1 of ARAS dataset. The algorithms are named with the convention of {EH-Pred}-{EM}.

Method	Pred. Horizon H	Year 2017	2018	2019	2020
MTCP	**H = 2**	**1.96**	**1.99**	**1.87**	**1.80**
	H = 3	**1.96**	**1.99**	**1.93**	**1.82**
	H = 4	**1.98**	**2.00**	**1.91**	**1.93**
	H = 5	**1.98**	**2.03**	**1.91**	**1.87**
MTCP-Random	H = 2	2.58	2.65	2.49	2.37
	H = 3	2.60	2.65	2.58	2.45
	H = 4	2.77	2.74	2.69	2.61
	H = 5	2.72	2.75	2.66	2.64
Ensemble-ML	H = 2	2.66	2.72	2.64	2.63
	H = 3	2.90	3.04	2.95	2.86
	H = 4	3.09	3.24	3.08	3.01
	H = 5	3.17	3.25	3.12	3.14
Ensemble-Random	H = 2	5.44	5.60	5.36	5.32
	H = 3	6.19	6.23	6.00	6.22
	H = 4	6.33	6.65	6.22	6.42
	H = 5	6.47	6.74	6.45	6.57
Point Prediction	H = 2	2.60	2.62	2.49	2.35
	H = 3	2.60	2.62	2.49	2.35
	H = 4	2.60	2.62	2.49	2.35
	H = 5	2.60	2.62	2.49	2.35

Table I presents a comparison of the MAE for EM decisions for all the EM algorithms from 2017-2020 for user 1 of the ARAS dataset. We note that other users show similar performance.

The table shows that the proposed MTCP approach demonstrates consistent performance with low MAE values across all horizons and years. For instance, the MAE for $H = 2$ ranges between 1.87 and 1.96 J across the four years, showing minimal variation. This highlights the robustness of our method when adjusting the prediction horizon. In contrast, baseline methods exhibit higher MAE values overall, with the Ensemble-Random approach showing the largest errors, ranging from 5.32 to 6.74 across horizons and years. The results indicate that our MTCP method consistently outperforms baseline approaches with different prediction horizons.

3) Evaluation of EM Decisions over a Day: This section analyzes the accuracy of the EM approaches against the optimal Oracle for a given user and day. Specifically, we obtain the energy decisions and battery energy levels over a day from all approaches and compare them to the optimal Oracle. Figure 4 shows energy allocations and battery energy levels in (a) and (b) for user four in ARAS dataset. It also shows energy allocations and battery energy levels in (c) and (d) for user six in Mannheim dataset.

As expected, MTCP maintains levels closer to Oracle, demonstrating its ability to make effective energy decisions. In contrast, baseline methods show larger deviations, with MTCP-Random and Point Prediction exhibiting the most significant mismatches. Overall, the figure illustrates that MTCP provides superior energy allocations and battery management compared to baseline approaches.

4) Comparison of Error with Different EM Approaches: EH patterns fluctuate over the year due to seasonal variations. Therefore, we demonstrate the effectiveness of our approach regardless of the season and the year. To this end, Figure 5 shows the average MAE and variance in each month across years. We can see that the MAE for ARAS dataset varies for different approaches with Ensemble-Random approach showing the highest error. In contrast, MTCP has the lowest MAE among all approaches. Similarly, MTCP has the lowest error for Mannheim dataset as well.

Next, we analyze the MAE across years. Specifically, Figures 6

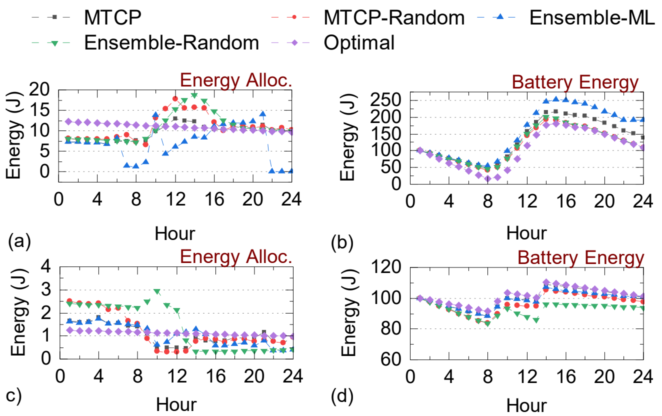

Fig. 4: Comparison of energy allocation and battery energy with MTCP, MTCP-Random, Ensemble-ML, Point Prediction, and Optimal for (a) and (b) user four in ARAS and (c) and (d) user six in Mannheim dataset for the day June 28th, 2017.

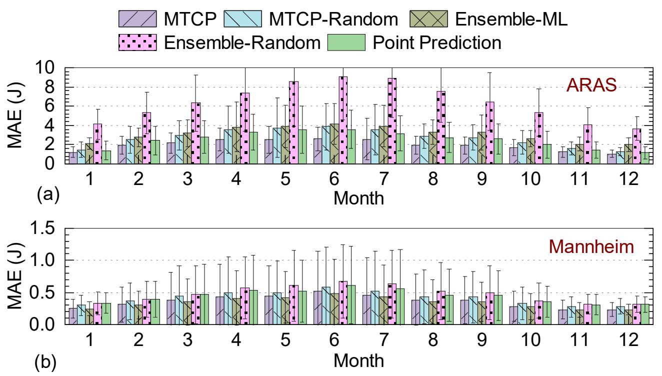

Fig. 5: Monthly Mean Absolute Error (Mean and standard deviation) of MTCP, MTCP-Random, Ensemble-ML, Ensemble-Random, and Point Prediction with user one in (a) ARAS (b) Mannheim dataset.

and 7 show the yearly MAE across different users in both datasets. The figures show the mean of MAE along with its standard deviation. MTCP consistently outperforms baseline approaches across all years, thus reflecting its ability to adapt effectively to changing EH patterns. These results underscore the robustness of MTCP in providing accurate predictions across diverse users and years.

E. Implementation Overhead

We employ the TI-CC2652R microcontroller [43] to measure the energy consumption and execution time of the proposed algorithms: MTCP, the IGP algorithm, and the ML model. The MTCP and ML models are implemented as multi-layer perceptrons. The MTCP algorithm includes constructing uncertainty regions based on neural network outputs. Measurements show that MTCP requires approximately 58 ms for execution, with an associated energy cost of 0.6 mJ per operation. The IGP algorithm, which iteratively optimizes energy usage, incurs a higher computational load due to its gradient-based approach. Execution of a single IGP operation requires approximately 300 ms, with an energy consumption of 8.6 mJ. Finally, the overhead measurement of the ML model shows that a single inference takes 188 ms, consuming 2.0 mJ of energy.

These overheads are modest in comparison to the operational timeframes and energy budgets of typical IoT applications. The overhead is included as part of the optimization in Equation 3 for all approaches. Moreover, the overhead can be minimized at runtime

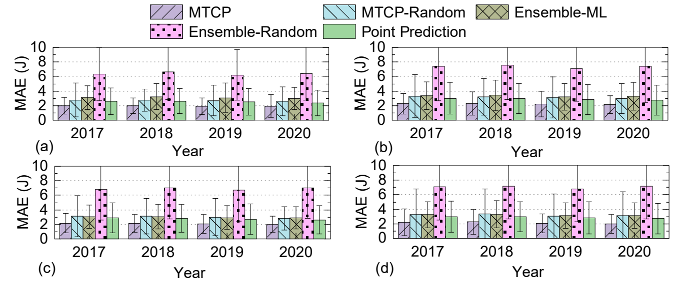

Fig. 6: Yearly Mean Absolute Error (Mean and standard deviation) of MTCP, MTCP-Random, Ensemble-ML, Ensemble-Random, and Point Prediction with different users in ARAS dataset.

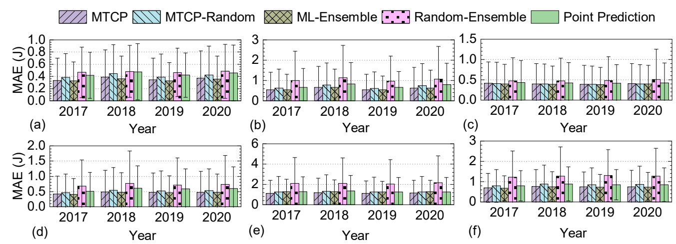

Fig. 7: Yearly Mean Absolute Error (Mean and standard deviation) of MTCP, MTCP-Random, Ensemble-ML, Ensemble-Random, and Point Prediction with different users in Mannheim dataset.

optimizing the Monte Carlo sampling and the IGP algorithm. In summary, the overhead of these algorithms remains within acceptable limits for resource-constrained IoT devices, ensuring their practical applicability in real-world scenarios.

VII. Conclusion

Wearable and IoT devices hold great potential for applications such as health monitoring and digital agriculture. Despite this promise, their adoption is often limited by the constraints of their small battery capacities. Ambient EH provides a viable solution to extend the battery life of these devices. However, the unpredictable nature of EH poses significant challenges for EM in wearable devices. Efficient EM relies on accurate predictions of future EH and the understanding of uncertainty associated with those predictions. Therefore, this paper proposed a novel energy management framework that leverages conformal prediction (CP) to provide uncertainty-aware EH forecasts over multiple prediction horizons. To enhance decision-making, an ML model is employed to select the optimal EM decision from multiple CP-generated uncertainty regions, balancing energy efficiency and device performance. The proposed approach outperforms baseline methods by ensuring reliable EM decisions, even under highly stochastic EH scenarios.

Acknowledgment

This work was supported in part by NSF CAREER award CNS-2238257 and in part by the AgAID AI Institute for Agriculture Decision Support, supported by the United States Department of Agriculture (USDA) - National Institute of Food and Agriculture (NIFA) award #2021-67021-35344.

References

[1] L. Atzori, A. Iera, and G. Morabito, "The Internet of Things: A Survey," *Computer Networks*, vol. 54, no. 15, pp. 2787–2805, 2010.

[2] A. J. Espay *et al.*, "Technology in Parkinson's Disease: Challenges and Opportunities," *Movt. Disorders*, vol. 31, no. 9, pp. 1272–1282, 2016.

[3] J.-F. Daneault, "Could Wearable and Mobile Technology Improve the Management of Essential Tremor?" *Frontiers in neurology*, vol. 9, pp. 257:1–257:8, 2018.

[4] D. Hussein, T. Belkhouja, G. Bhat, and J. R. Doppa, "Reliable Machine Learning for Wearable Activity Monitoring: Novel Algorithms and Theoretical Guarantees," in *Proc. Int. Conf. on Comput.-Aided Design*, 2022, pp. 33:1–33:9.

[5] D. Hussein, T. Belkhouja, G. Bhat, and J. Doppa, "Sensor-Aware Data Imputation for Time-Series Machine Learning on Low-Power Wearable Devices," *ACM Trans. Design Autom. Electr. Syst.*, vol. 30, no. 1, pp. 1–27, 2025.

[6] D. Hussein, T. Belkhouja, G. Bhat, and J. R. Doppa, "Energy-Efficient Missing Data Recovery in Wearable Devices: A Novel Search-Based Approach," in *Int. Symp. on Low Power Electron. and Des. (ISLPED)*, 2023, pp. 1–6.

[7] A. Kansal, J. Hsu, S. Zahedi, and M. B. Srivastava, "Power Management in Energy Harvesting Sensor Networks," *ACM Trans. Embedd. Comput. Syst.*, vol. 6, no. 4, p. 32, 2007.

[8] M. Odema, N. Rashid, and M. A. Al Faruque, "Energy-Aware Design Methodology for Myocardial Infarction Detection on Low-Power Wearable Devices," in *Proc. Asia and South Pacific Des. Autom. Conf.*, 2021, pp. 621–626.

[9] Y. Tuncel, S. Bandyopadhyay, S. V. Kulshrestha, A. Mendez, and U. Y. Ogras, "Towards Wearable Piezoelectric Energy harvesting: Modeling and Experimental Validation," in *Int. Symp. on Low Power Electron. and Des. (ISLPED)*, 2020, pp. 55–60.

[10] D. Hussein, N. Yamin, and G. Bhat, "EMI: Energy Management Meets Imputation in Wearable IoT Devices," *IEEE Trans. Comput.-Aided Design Integr. Circuits Syst.*, vol. 43, no. 11, pp. 3792–3803, 2024.

[11] N. Yamin and G. Bhat, "Indoor–Outdoor Energy Management for Wearable IoT Devices With Conformal Prediction and Rollout," *IEEE Trans. Comput.-Aided Design Integr. Circuits Syst.*, vol. 43, no. 11, pp. 3370–3381, 2024.

[12] Y. Romano, E. Patterson, and E. Candes, "Conformalized Quantile Regression," *Advances in Neural Information Processing Systems*, vol. 32, 2019.

[13] H. Alemdar, H. Ertan, O. D. Incel, and C. Ersoy, "ARAS Human Activity Datasets in Multiple Homes with Multiple Residents," in *Int. Conf. Perv. Comput. Tech. Health. and Work.*, 2013, pp. 232–235.

[14] T. Sztyler, J. Carmona, J. Völker, and H. Stuckenschmidt, "Self-tracking Reloaded: Applying Process Mining to Personalized Health Care from Labeled Sensor Data," in *Trans. Petri Nets and other Models Concur.* Springer, 2016, pp. 160–180.

[15] N. Yamin and G. Bhat, "Uncertainty-Aware Energy Harvest Prediction and Management for IoT Devices," *ACM Trans. Design Autom. Electr. Syst.*, vol. 28, no. 5, pp. 1–33, 2023.

[16] A. Cammarano, C. Petrioli, and D. Spenza, "Pro-Energy: A Novel Energy Prediction Model for Solar and Wind Energy-Harvesting Wireless Sensor Networks," in *Int. Conf. on Mobile Ad-Hoc and Sensor Syst.*, 2012, pp. 75–83.

[17] J. R. Piorno, C. Bergonzini, D. Atienza, and T. S. Rosing, "Prediction and Management in Energy Harvested Wireless Sensor Nodes," in *Int. Conf. Wireless Comm., Vehicular Tech., Info. Theory and Aerospace & Electron. Syst. Tech.*, 2009, pp. 6–10.

[18] Z. Wan, Y. Tan, and C. Yuen, "Review on Energy Harvesting and Energy Management for Sustainable Wireless Sensor Networks," in *2011 IEEE 13th Int. Conf. on Communication Technology*, 2011, pp. 362–367.

[19] M.-H. Roy and D. Larocque, "Prediction Intervals with Random Forests," *Statistical Methods in Medical Research*, vol. 29, no. 1, pp. 205–229, 2020.

[20] D. L. Shrestha and D. P. Solomatine, "Machine Learning Approaches for Estimation of Prediction Interval for the Model Output," *Neural Networks*, vol. 19, no. 2, pp. 225–235, 2006.

[21] G. Narang, R. Ayoub, M. Kishinevsky, J. R. Doppa, and P. P. Pande, "Uncertainty-Aware Online Learning for Dynamic Power Management in Large Manycore Systems," in *Int. Symp. on Low Power Electron. and Des. (ISLPED)*, 2023, pp. 1–6.

[22] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*. Springer, 2005, vol. 29.

[23] Y. Romano, M. Sesia, and E. Candes, "Classification with Valid and Adaptive Coverage," *Advances in Neural Inf. Processing Sys.*, vol. 33, pp. 3581–3591, 2020.

[24] I. Gibbs, J. J. Cherian, and E. J. Candès, "Conformal Prediction with Conditional Guarantees," *arXiv preprint arXiv:2305.12616*, 2023.

[25] R. J. Tibshirani, R. Foygel Barber, E. Candes, and A. Ramdas, "Conformal Prediction under Covariate Shift," *Advances in Neural Information Proc. Syst. (NeurIPS)*, vol. 32, 2019.

[26] Q. Regression, *Handbook of Quantile Regression*. CRC Press: Boca Raton, FL, USA, 2017.

[27] R. Koenker and G. Bassett Jr, "Regression Quantiles," *Econometrica: Journal of the Econometric Society*, pp. 33–50, 1978.

[28] S. Ghosh, T. Belkhouja, Y. Yan, and J. R. Doppa, "Improving Uncertainty Quantification of Deep Classifiers via Neighborhood Conformal Prediction: Novel Algorithm and Theoretical Analysis," in *Proc. of AAAI Conf.*, 2023, pp. 7722–7730.

[29] S. Ghosh, Y. Shi, T. Belkhouja, Y. Yan, J. Doppa, and B. Jones, "Probabilistically Robust Conformal Prediction," in *UAI Conf.*, ser. Proc. of Machine Learning Research, vol. 216. PMLR, 2023, pp. 681–690.

[30] Y. Shi, S. Ghosh, T. Belkhouja, J. Doppa, and Y. Yan, "Conformal Prediction for Class-wise Coverage via Augmented Label Rank Calibration," in *Advances in Neural Information Processing Sys. (NeurIPS)*, 2024.

[31] H. Shahrokhi, D. R. Roy, Y. Yan, V. Arnaoudova, and J. R. Doppa, "Conformal Prediction Sets for Deep Generative Models via Reduction to Conformal Regression," 2025. [Online]. Available: https://arxiv.org/abs/2503.10512

[32] C. M. Vigorito, D. Ganesan, and A. G. Barto, "Adaptive Control of Duty Cycling in Energy-Harvesting Wireless Sensor Networks," in *Proc. Conf. on Sensor, Mesh and Ad Hoc Comm. and Networks*, 2007, pp. 21–30.

[33] T. Basaklar, Y. Tuncel, and U. Y. Ogras, "tinyMAN: Lightweight Energy Manager using Reinforcement Learning for Energy Harvesting Wearable IoT Devices," in *tinyML Research Symp.*, 2022, pp. 1–7.

[34] M. Geisler *et al.*, "Human-Motion Energy Harvester for Autonomous Body Area Sensors," *Smart Materials and Structures*, vol. 557, no. 1, p. 012024, 2017.

[35] G. Bhat, J. Park, and U. Y. Ogras, "Near-Optimal Energy Allocation for Self-Powered Wearable Systems," in *Proc. Int. Conf. on Comput.-Aided Design*, 2017, pp. 368–375.

[36] N. Yamin, G. Bhat, and J. R. Doppa, "DIET: A Dynamic Energy Management Approach for Wearable Health Monitoring Devices," in *2022 Des., Autom. & Test in Europe Conf. & Exhibition (DATE)*, 2022, pp. 1365–1370.

[37] D. Hussein, G. Bhat, and J. R. Doppa, "Adaptive Energy Management for Self-Sustainable Wearables in Mobile Health," in *Proc. AAAI Conf.*, vol. 36, no. 11, 2022, pp. 11 935–11 944.

[38] T. Huynh, M. Fritz, and B. Schiele, "Discovery of Activity Patterns using Topic Models," in *Proc. of the 10th Int. Conf. on Ubiquitous Computing*, 2008, pp. 10–19.

[39] S. P. Boyd and L. Vandenberghe, *Convex Optimization*. Cambridge University Press, 2004.

[40] N. Karakoç *et al.*, "Multi-Layer Decomposition of Network Utility Maximization Problems," *IEEE/ACM Trans. on Netw.*, vol. 28, no. 5, pp. 2077–2091, 2020.

[41] FlexSolarCells, "SP3-37 Datasheet," 2013, https://bit.ly/3dcJ0lK, accessed 3/28/2021.

[42] A. Andreas and T. Stoffel, "NREL Solar Radiation Research Laboratory (SRRL): Baseline Measurement System (BMS); Golden, Colorado (Data); NREL Report No. DA-5500-56488," 1981, accessed 3/20/2023.

[43] Texas Instruments Inc., "CC2652R Microcontroller," [Online] https://www.ti.com/product/CC2652R, accessed August 12, 2022, 2018.